\CurrentAudience: **Fac**

PH113 / Assignment - Module IV: Quantum Physics

(Dated: March 11, 2024)

1. Calculate the de Broglie wavelength of (i) a 10 eV electron, and (ii) a 1.0MeV neutron. (Assume classical non-relativistic particles; Mass of neutron is 1.675×10^{-27} kg = 939.6 Mev/c^2 $\approx m_{\text{proton}}$.)

   Solution. Considering:

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}},$$

   one gets: (i) 3.9 Å, and (ii) 286.2 Å.

> *Comment: For relativistic particles, one can show that:*
>
> $$\lambda = \frac{hc}{\sqrt{E_K^2 + 2m_0 E_K c^2}}$$
>
> *where E_K is the kinetic energy.*

2. An increase of 150 eV in the energy of an electron reduces its de Broglie wavelength to half of its earlier value. Calculate the initial de Broglie wavelength.

   Solution. We know that $\lambda = h/\sqrt{2mE}$. Inverting this relation, we obtain:

$$\begin{aligned}
E_0 &= \frac{1}{2m}\frac{h^2}{\lambda_0^2} \\
\text{and, } E_1 &= \frac{1}{2m}\frac{h^2}{\lambda_1^2} \\
\therefore E_0 + \Delta E &= \frac{1}{2m}\frac{4h^2}{\lambda_0^2} \\
\implies \Delta E &= \frac{1}{2m}\frac{3h^2}{\lambda_0^2} \\
\implies \lambda_0 &= \sqrt{\frac{3h^2}{2m\Delta E}}
\end{aligned}$$

Pluggin in the values, we get:

$$\begin{aligned}\lambda_0 &= \sqrt{\frac{3\times(4.136)^2\times10^{-30}\times3\times10^8}{2\times0.511\times10^6\times150}} \qquad [\mathrm{m}]\\ &= 1.736\text{Å}\end{aligned}$$

3. What is the potential through which a proton must be accelerated so that its de Broglie wavelength is 2.0 Å?

Solution. The energy of a particle with charge e accelerated through a potential V is $E = eV$. Therefore, to have a de Broglie wavelength of $\lambda = 2.0$ Å, we have:

$$\begin{aligned}V &= \frac{1}{2me}\frac{h^2}{\lambda^2}\\ &= \frac{(6.63)^2}{2\times1.67\times1.6\times4}\times10^{-68+19+20+27} \qquad [\mathrm{V}]\\ &= 20.5\ \mathrm{mV}\end{aligned}$$

4. An electron has a de Broglie wavelength equal to that of a photon of red light having wavelength 620 nm. (i) Calculate the energy of the electron. (ii) What would its de Broglie wavelength be if the energy of the electron were equal to the energy of this photon?

Solution. (i) The energy of the electron with a de Broglie wavelength $\lambda = \lambda_{\mathrm{ph}} = 620\,\mathrm{nm} = 6.2\times10^{-7}\,\mathrm{m}$ is given by:

$$\begin{aligned}E &= \frac{1}{2m}\frac{h^2}{\lambda_{\mathrm{ph}}^2}\\ &= 6.28\times10^{-21} \quad \mathrm{J}\\ &\approx 0.04 \quad \mathrm{eV}\end{aligned}$$

(ii) The energy of the photon of wavelength λ_{ph} is:

$$E_{\mathrm{ph}} = \frac{hc}{\lambda_{\mathrm{ph}}} = 2.0013\,\mathrm{eV}$$

The de Broglie wavelength of an electron with energy equal to E_{ph} would be given by:

$$\lambda = \frac{h}{\sqrt{2mE_{\mathrm{ph}}}} = 2.9\times10^{-12} \quad \mathrm{eV}$$

5. **Compton effect.** X-rays of wavelength 10.0 pm are scattered through a target.

   (a) Find the wavelength of the x-rays scattered through 45°.

   (b) Find the maximum wavelength present in the scattered x-rays.

   (c) Find the maximum kinetic energy of the recoil electrons.

Solution. page 74, Beiser 4th Ed.

(a) We have:

$$\lambda' - \lambda = \lambda_c(1 - \cos\phi)\,.$$

Therefore, using $\phi = 45°$ and $\lambda = 10\,\text{pm}$, we get

$$\begin{aligned}\lambda' &= 10\,\text{pm} + 0.293\lambda_c \\ &= 10.7\,\text{pm}\end{aligned}$$

where we have used $\lambda_c = 2.426\,\text{pm}$ for electrons.

> *Comment: The Compton wavelength $\lambda_c = h/m_0c$ for other heavier particles is even smaller than that for an electron.*

(b) The maximum wavelength of the scattered x-rays corresponds to largest possible value of $\lambda' - \lambda$, which is realized for $\phi = 180°$ such that $(1 - \cos\phi) = 2$. Therefore, one finally obtains:

$$\lambda' - \lambda = 2\lambda_c\,,$$

leading to $\lambda' = 14.9\,\text{pm}$.

(c) The max recoil kinetic energy is equal to difference between the incident and scattered photon energies (max energy transferred to particle). Therefore,

$$\begin{aligned}K_{\text{max}} &= h(\nu - \nu') \\ &= h(\frac{1}{\lambda} - \frac{1}{\lambda'})\,.\end{aligned}$$

Using λ' from part (b) above, we get $K_{\text{max}} = 40.8\,\text{keV}$.

6. What is the frequency of an x-ray photon whose momentum is 1.1×10^{-23} kg m/s^2?

Solution. The energy of the photon is given by:

$$\nu = \frac{pc}{h} = 5.0 \times 10^{18}\,\mathrm{Hz}\,.$$

7. How much energy must a photon have if it is to have a momentum of 10 MeV proton?

Solution. Using $E = mv^2/2$ for the proton, we get

$$v \;=\; 4.38 \times 10^{7}\,\mathrm{m/s}$$

therefore,

$$p \;=\; mv = 7.31 \times 10^{-20}\,\mathrm{kg\ m/s}\,.$$

Using this value of momentum for the photon, the energy is given by:

$$E = pc = 2.19 \times 10^{-11}\,\mathrm{J} = 137\,\mathrm{MeV}\,.$$

8. An x-ray photon of initial frequency 3×10^{19} Hz collides with an electron and is scattered through 90°. Find its new frequency.

Solution. Rewriting the equation for the Compton scattering in terms of frequencies, together with $\phi = 90°$ ($\Longrightarrow \cos\phi = 0$), we have

$$\frac{c}{\nu'} = \frac{c}{\nu} + \lambda_{\mathrm{c}}$$

Solving for ν',

$$\nu' = \left[\frac{1}{\nu} + \frac{\lambda_{\mathrm{c}}}{c}\right]^{-1} = 2.4 \times 10^{19}\,\mathrm{Hz}\,.$$

9. An x-ray photon with initial frequency of 1.5×10^{19} Hz emerges from a collision with an electron with a frequency of 1.2×10^{19} Hz. How much kinetic energy was imparted to the electron?

Solution. Assuming elastic collision (and that the electron is initially at rest), the kinetic energy of the electron is given by:

$$E_k^e = E_f^p - E_i^p \,,$$

where, $E_{f/i}^p$ is the final/initial energy of the photon. Plugging in the values, we get

$$\begin{aligned} E_k^e &= 0.3 \times h \times 10^{19} \\ &= 1.24 \times 10^4 \, \mathrm{eV} \,, \end{aligned}$$

10. At what scattering angle will incident 100 keV x-rays leave a target with an anergy of 90 keV?

Solution. Solving the Compton equation for $\cos\phi$, we get:

$$\begin{aligned} \cos\phi &= 1 + \frac{\lambda}{\lambda_c} - \frac{\lambda'}{\lambda_c} \\ &= 1 + \frac{m_0 c^2}{E} - \frac{m_0 c^2}{E'} = 0.432 \\ \Longrightarrow \phi &= 64^\circ \,, \end{aligned}$$

where we used $E = hc/\lambda$ and $\lambda_c = h/m_0 c$.

11. A photon of frequency ν is scattered by an electron initially at rest. Verify that the maximum kinetic energy of the recoil electron is

$$K_{\text{max}} = \frac{2h^2\nu^2}{m_0 c^2} \frac{1}{1 + 2h\nu/m_0 c^2}$$

Solution. Hint: use $\cos\phi = -1$ and expression for kinetic energy.

12. **Davison-Germer expt.** What effect on the scattering angle in theDavisson-Germer experiment does increasing theelectron energy have?

Solution. Increasing the electron energy increases the electron's momentum, and hence decreases the electron's de Broglie wavelength. A smaller de Broglie wavelength results in a smalle r scattering angle (see Beiser).

13. In the Davisson-Germer experiment, the energy of the electron entering the crystal increases, which reduces its de Broglie wavelength. Consider a beam of 54 eV electrons directed at a nickel target. The potential energy of an electron that enters the target changes by 26 eV. (a) Compare the electron speeds outside and inside the target. (b) Compare the respective de Broglie wavelengths.

    Solution. For the given energies, a non-relativistic calculation is sufficient. We have

    $$v = \sqrt{\frac{2E}{m}} = \sqrt{\frac{2 \times 54 \times 1.6}{9.1}} \times 10^{(-19+31)/2} = 4.36 \text{ [m/s]}$$

    outside the crystal, and (from a similar calculation, with K = 80eV [shift of 26 eV]),$v = 5.30x10^6$ m/s inside the crystal (keeping an extra significant figure in both calculations).

    (b) With the speeds found in part (a), the de Brogile wavelengths are found from

    $$\lambda = \frac{h}{p} = \frac{h}{mv} = 1.67 \times 10^{-10}\text{m}$$

    or 0.167 nm outside the crystal. Similar calculations leads to 0.137 nm inside the crystal.

14. Write down the Schrödinger equation for free particle and its solution.

15. **Particle in a box.** Obtain the energy values and wavefunctions of a quantum particle in a one-dimensional box. (Write down the Schrödinger equation for particle in a box and solve it). Plot the lowest two energy wavefunctions.

16. The lowest energy of a quantum particle trapped in a (unspecified) box is 1 eV. What are the next two higher energies the particle can have.

    Solution. Assuming 1D box, we can write the energies as:

    $$E_n = n^2 \frac{h^2}{8mL^2} \qquad n = 1, 2, 3 \dots .$$

    Therefore, $E_2 = 4E_1 = 4\,\text{eV}$ and $E_3 = 9E_1 = 9\,\text{eV}$.

17. Find the energy levels of an electron is a box 0.1 nm wide.

    Solution. page 105, Beiser 4th Ed

    $$E_n = n^2 \frac{h^2}{8mL^2} = 38n^2\,\text{eV}\,.$$

18. Find the energy levels of a 10 g marble in a box 10 cm wide assuming the marble to be a quantum particle.

> *Comment: Using this example, convince yourself that quantum effects from a non-quantum object is actually negligible.*

Solution. With $m = 10\text{g} = 10^{-2}$ kg and $L = 10$ cm $= 10^{-1}$ m, we get:

$$\begin{aligned} E_n &= \frac{n^2 h^2}{8mL^2} \\ &= \frac{n^2\,(6.626 \times 10^{-34})^2}{8\,10^{-2}\,(10^{-1})^2} \\ &= 5.5 \times 10^{-64} n^2 \text{ J.} \end{aligned}$$

> *Comment: The above value is the minimum energy the marble can have. A marble with this kinetic energy has a speed of only* 3.3×10^{-31} *m/s and, therefore, cannot be experimentally distinguished from a stationary particle (marble).*
>
> *Considering a modest speed of 1/3 m/s corresponds to* $n = 10^{30}$. *At such large values of* n, *the energy level spacing is so tiny that is impossible to say with confidence whether the marble will take only (quantum) energy values or any energy whatever.*

19. Obtain the expression for the energy levels (in MeV) of a neutron confined to a one-dimensional box 10^{-14} m wide. What is the neutron's minimum energy?

> *Comment: The diameter of an atomic nucleus is of this order of magnitude.*

Solution. Energy for a quantum particle in a one-dimensional box is given by:

$$\begin{aligned} E_n &= n^2 \frac{h^2}{8mL^2} \\ &= 20.5\, n^2\, \text{MeV}\,. \end{aligned}$$

20. **Uncertainty principle.** A measurement establishes the position of a proton with an accuracy of $\pm 10^{-11}$ m. Find the uncertainty in the proton's position 1 s later. Assume $v << c$.

Solution. Let Δx_0 denote the position uncertainty at $t = 0$. Therefore, the corresponding undercertainty in momentum (at $t = 0$) is given by:

$$\Delta p_0 \geq \frac{\hbar}{2\Delta x_0}\,.$$

Since $v << c$, the momentum uncertainty is $\Delta p = \Delta(mv) = m_0 \Delta v$ and the uncertainty in proton's velocity is:

$$\begin{aligned}\Delta v_0 &= \frac{\Delta p}{m_0} \\ &\geq \frac{\hbar}{2m_0\Delta x_0}\,.\end{aligned}$$

The distance x travelled by the particle in time t cannot be known more accurately than:

$$\begin{aligned}\Delta x &= t\Delta v_0 \\ &\geq \frac{t\hbar}{2m_0\Delta x_0}\,.\end{aligned}$$

Evidently, Δx is inversely proportional to Δx_0, that is the more precisely we know about the particle at any instance of time, the less we know about its later position at time t.

Plugging in the values, we get:

$$\Delta x \geq 3.15 \times 10^3 m.$$

21. Discuss the Plank's theory of black-body radiation.

Comment: ***Concluding Remarks.*** *The students are* ***strongly recommended*** *to take a look at the relevant solved examples from the book Concepts of Modern Physics by Arthur Beiser, and also solve them explicitly.*

It is, in general, not possible to include all those examples here and discuss them explicitly. Based *on the lectures, the students are expected to be have sufficient skills to solve those problems. The relevant topics can be found from the course syllabus.*